

Christof Teuscher

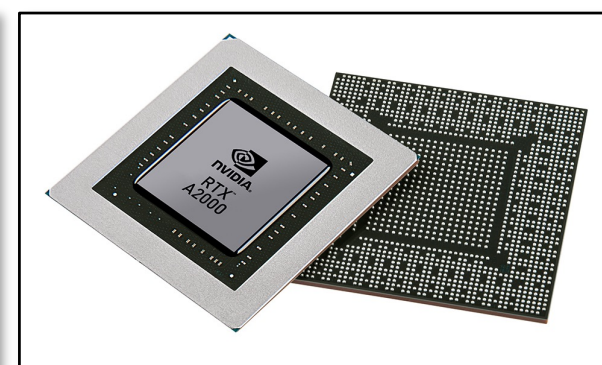
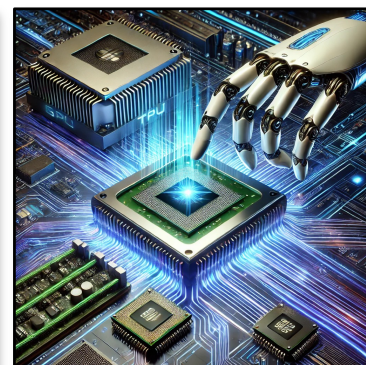
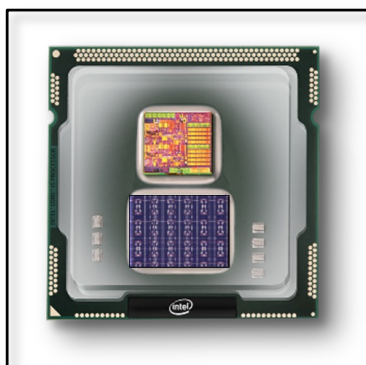
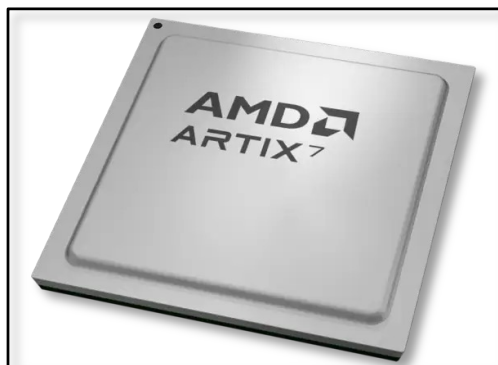
ECE 410/510: Hardware for AI and ML, Spring 2026

Week 6: Recap, outlook, and reminders

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What did you learn last week?

Recap

- No free lunch theorem (no algorithm consistently outperforms random search or any other algorithm).
- BF16
- TPU (matrix multiplication + activation unit)
- Systolic array
- Mapping a neural net on a systolic array
- Transformers (encoder, decoder): main operations

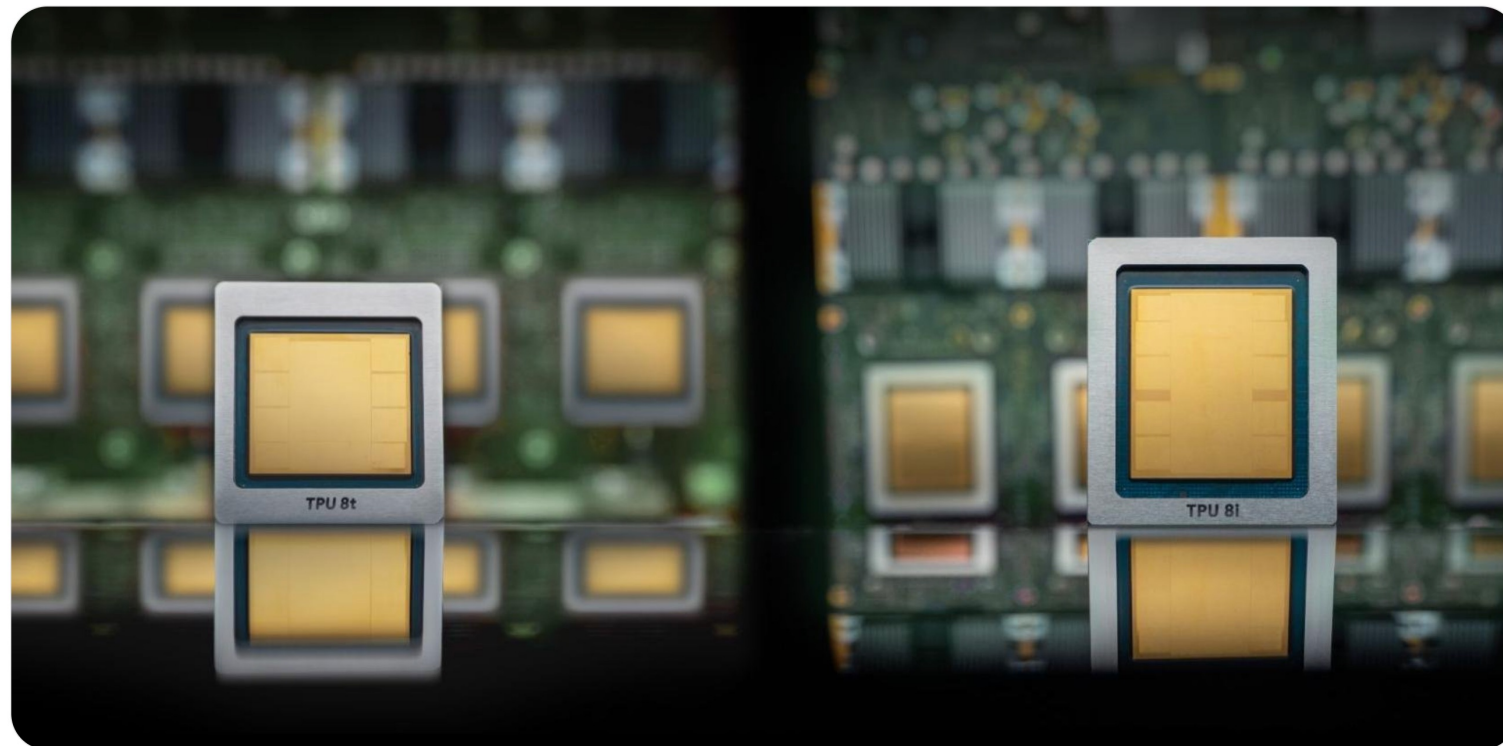
Quiz 1 is open until Wed, 11:59pm

- Do not wait until the last minute!
- Talking about what you know is extremely important. *“If you can't explain it simply, you don't understand it well enough.”*
- It's a skill that can and should be developed.
- One goal in this course is to develop and build that “muscle.”
- When you graduate, you want the actual knowledge, not just a piece of paper.



Inside the eighth-generation TPU: An architecture deep dive

April 22, 2026



<https://cloud.google.com/blog/products/compute/tpu-8t-and-tpu-8i-technical-deep-dive>

Google TPU 8: What's New?

TPU 8t — Pre-Training

SparseCore offloads embedding lookups from MXU

Native FP4 — doubles MXU throughput (12.6 PFLOPs)

Virgo Network: 4x DCN bandwidth; scales to 1M+ chips

TPUDirect Storage: 10x faster I/O vs. Ironwood

2.7x training price-performance vs. Ironwood

TPU 8i — Inference & Serving

3x more SRAM — full KV cache on-chip for long-context

CAE: 5x lower collective latency for autoregressive decoding

Boardfly topology: 7 hops vs. 16 (3D torus) — 50% less latency

Optimized for MoE & reasoning; 1,152 chips fully connected

80% inference price-performance vs. Ironwood

Both chips: Arm Axion CPU • Native FP4 • JAX / PyTorch / XLA • 2x perf/watt vs. Ironwood

TPU 8t vs. TPU 8i: Comparison

Feature	TPU 8t	TPU 8i
Workload	Large-scale pre-training	Inference, serving, reasoning
Network	3D torus (9,600 chips/pod)	Boardfly (max 7 hops, 1,152 chips)
Special HW	SparseCore (embeddings)	CAE (collective acceleration)
HBM / SRAM	216 GB / 128 MB	288 GB / 384 MB (3x SRAM)
Peak FP4	12.6 PFLOPs	10.1 PFLOPs
HBM BW	6,528 GB/s	8,601 GB/s (~1.3x)
vs. Ironwood	2.7x training price-perf	80% inference price-perf

- Pre-training is the first and most computationally expensive phase of building a large AI model.
- TPU 7 consumed about 600W.

Senior AI Software Architect - Runtime

Apply



Intel Careers

US, Oregon, Hillsboro
US, California, Santa Clara
US, Texas, Austin

Full time
Posted 5 Days Ago
JR0283399

Job Details:

Job Description:

For nearly a decade, Intel's Neuromorphic Computing Lab-together with a global ecosystem of 250+ research groups-has explored architectures, algorithms, and software inspired by the brain's extraordinary efficiency, scalability, and adaptability. Our Loihi series of research chips pioneered event-driven, sparse, and massively parallel neuro-inspired processing, fueling over 100 peer-reviewed publications that validate the promise of this novel approach.

Now, we're entering an exciting new chapter: transforming these breakthroughs into real-world products that will power the coming era of physical AI systems-beyond the reach of GPUs and mainstream AI accelerators.

If you are passionate about pushing the boundaries of computing, from transistor-level innovation to software abstractions, join us. Help define the next wave of AI technology that harnesses the proven advantages of Intel's neuromorphic computing technology with the versatility demanded by modern AI workloads.

Position Overview:

Join the team building Intel's product-grade neuromorphic AI execution stack targeting edge and robotic systems. In this role, you will architect and lead the development of the firmware, runtime, and performance infrastructure, working closely with hardware, compiler, systems, and application teams to deliver a robust, observable, and high-performance software stack. You will set technical direction for a critical part of the platform while remaining hands-on in architecture and implementation.

Responsibilities:

- Define and drive the technical vision, architecture, and roadmap for Intel's neuromorphic runtime software stack, while contributing hands-on to critical design and implementation.
- Architect, develop, and optimize firmware and runtime components, including execution control, scheduling, data movement, memory management, and device interfaces.
- Architect and deliver performance infrastructure, including profiling, tracing, debugging, benchmarking, and regression frameworks needed for optimization and product release.
- Integrate the runtime stack into existing robotics and edge software ecosystems, to enable efficient deployment and developer adoption.
- Collaborate with hardware, compiler, systems, application, and solution teams, as well as customers and external partners, to translate platform and workload requirements into robust software capabilities.
- Provide technical leadership across the runtime area by establishing engineering practices, driving software quality and release discipline, mentoring engineers, and helping scale the team over time.

Qualifications:

Minimum Qualifications

- BSc, MSc, or PhD in Computer Science, Electrical/Computer Engineering, Physics, Mathematics, or a related technical field.
- 5+ years of experience developing firmware, runtime, performance, or other low-level systems software for AI or accelerators, including execution environments, profiling, debugging, and related infrastructure, with experience in technologies such as CUDA, LLVM, oneAPI, SYCL, ONNX, IREE, OpenVINO, TVM.
- 5+ years of experience leading software architecture for complex AI, runtime, or accelerator software systems, including defining technical direction and guiding implementation.
- 8+ years of experience developing maintainable, production-grade software in C/C++ and Python, including systems-level and performance-critical code, using disciplined engineering practices such as benchmarking, correctness and performance regression testing, build/toolchain management, and CI/CD.
- 5+ years of experience with AI, deep learning, or optimization workloads using frameworks such as PyTorch, JAX, or TensorFlow.

Preferred Qualifications

- Experience managing and mentoring engineers, and leading delivery across cross-functional teams spanning hardware, systems software, compiler, and applications.
- Experience with robotics, edge, or real-time systems, including software ecosystems such as ROS2.
- Experience in HW/SW co-design, hardware simulator development, or software bring-up for emerging accelerator architectures.
- Experience with parallel computing paradigms or spatial / unconventional AI hardware accelerators.
- Experience with embedded or real-time Linux, device-driver / platform bring-up, or other low-level deployment software for edge systems.
- Prior contributions to open-source software projects.



This week

Monday

- Ways to accelerate an algorithm
- Transformers: to spike or not to spike
- In-memory computing

Wednesday

- Codefest #6